

Risk-Based Allocation Strategies

Risk Parity, ERC, Hierarchical Clustering & HRP

*Politecnico di Milano — Financial Engineering
Buy Side — Laboratory 4b — FY 2025/2026*

01

Why Risk-Based?

Moving beyond mean–variance: managing what we can measure

Markowitz's mean–variance framework requires expected returns and a covariance matrix. Returns are notoriously hard to estimate, and tiny errors blow up the optimal weights. Risk-based strategies give up on returns and focus solely on what we can measure: risk.

From Markowitz to Risk-Based

Why practitioners abandon expected returns in portfolio construction

Limits of Mean–Variance

Tiny estimation errors → extreme weights

MVO needs μ and Σ as inputs. Expected returns are the dominant source of error and produce unstable, concentrated allocations.

The Risk-Based Idea

Estimate Σ only — diversify risk, not returns

Drop the return forecast entirely. Use the covariance structure to build portfolios that spread risk across assets, not capital. The result is more robust, less noisy, and easier to defend out-of-sample.

Risk Concentration in the 60/40 Portfolio

60% Equity / 40% Bonds looks diversified by capital — but equities carry far higher volatility. Once you weight by σ , ~90% of total portfolio risk sits in the equity bucket. Risk-based methods aim to fix exactly this kind of imbalance.

Key insight: capital diversification is not risk diversification.

02

Naive Risk Parity

The simplest correlation-blind risk-balancing rule

Inverse-volatility (IV) weighting allocates capital in proportion to $1/\sigma_i$. It is the simplest example of a risk-parity strategy — closed-form, fast, and surprisingly hard to beat.

Inverse Volatility (IV)

Each asset receives capital inversely proportional to its volatility

The Weighting Rule

Closed-form, no optimization required

$$w_i = (1/\sigma_i) / \sum_j (1/\sigma_j)$$

Each weight is just the reciprocal of a volatility forecast, then normalized so that the weights sum to one.

Why It Works in Practice

Stable, robust, and benchmark-beating in many regimes

Requires only marginal volatilities — no covariances, no inversion, no optimizer. The recipe is fast, transparent, and naturally tilts the portfolio away from the most volatile names.

Underlying Assumption & Main Drawback

IV implicitly assumes that the asset returns are uncorrelated (Σ diagonal). Under this assumption, equal capital scaled by $1/\sigma$ delivers equal risk per asset. When correlations are non-trivial — as in most equity universes — IV ignores them and can over-weight a cluster of correlated low-vol names, creating hidden risk concentration.

Key insight: IV is correlation-blind — robust, but risks hidden cluster concentration.

03

Equal Risk Contribution

Each asset contributes equally to total portfolio risk

Maillard, Roncalli & Teiletche (2010) proposed ERC: weights are chosen so that every asset accounts for the same share of $\sigma(w)$. Correlation-aware, well-defined, but no closed form — it requires solving a non-linear optimization problem.

Risk Contribution Decomposition

Euler's theorem makes $\sigma(w)$ additive across assets

1 Portfolio Volatility

$$\sigma(w) = \sqrt{w^T \Sigma w}$$

Volatility is a homogeneous function of degree 1 in w . Euler's theorem then gives a clean linear decomposition into per-asset risk contributions.

2 Marginal Risk Contribution

$$MRC_i = \partial \sigma(w) / \partial w_i = (\Sigma w)_i / \sigma(w)$$

MRC_i tells us how much portfolio risk changes per unit increase in w_i . It depends on covariances with every other asset.

3 Total Risk Contribution

$$RC_i = w_i \cdot (\Sigma w)_i / \sqrt{w^T \Sigma w}$$

Multiplying weight by marginal contribution gives the absolute risk each asset adds. By Euler, $\sum_i RC_i = \sigma(w)$.

4 ERC Condition

$$RC_i = RC_j \text{ for all } i, j \Rightarrow RC_i = \sigma(w)/N$$

All assets carry the same share $1/N$ of total risk. This is the defining condition of the equal-risk-contribution portfolio.

Solving for the ERC Portfolio

A non-linear program, with IV as a clean special case

The Optimization Problem

Equalize risk contributions via least-squares

$$\min_{\mathbf{w}} \sum_i \sum_j (\mathbf{w}_i (\Sigma \mathbf{w})_i - \mathbf{w}_j (\Sigma \mathbf{w})_j)^2$$

$$\text{s.t. } \sum_i w_i = 1, \quad w_i \geq 0$$

No closed form — requires a numerical solver and a good covariance estimate.
Inheriting estimation noise from Σ is the main vulnerability.

When IV = ERC

Σ diagonal $\Rightarrow$ closed-form coincidence

If asset returns are uncorrelated, the ERC optimum reduces algebraically to $w_i \propto 1/\sigma_i$ — the inverse-volatility rule. So IV is the correlation-free limit of ERC, which collapses gracefully when no correlation information is available.

Why ERC Sits Between IV and MVO

ERC keeps the spirit of risk parity (equalize risk, not capital) while accounting for correlations. It has no minimum-variance pretensions — but is much more stable than MVO because it ignores expected returns and avoids inverting Σ for the weights.

Key insight: correlation-aware risk parity — robustness with realistic structure.

04

Hierarchical Clustering

Turning a correlation matrix into a tree of asset groups

Agglomerative clustering merges assets bottom-up using a correlation-derived distance and a linkage rule. The output is a dendrogram that exposes the natural groupings hiding inside Σ .

From Correlation to Distance

Correlation is not a metric — but a simple transform makes it one

Correlation Distance

$$d_{ij} = \sqrt{0.5 \cdot (1 - \rho_{ij})}$$

Maps $\rho \in [-1, 1]$ into $d \in [0, 1]$: perfectly correlated assets sit at distance 0, perfectly anti-correlated at distance 1.

Satisfies the three metric axioms (non-negativity, symmetry, triangle inequality) — derivable from the L^2 norm of standardized returns.

Linkage Rules

How to define distance between two clusters

- Single linkage: minimum pairwise distance (chains)
- Complete linkage: maximum pairwise distance (compact clusters)
- Average linkage: mean pairwise distance (balanced)
- Ward linkage: minimizes within-cluster variance
- ...

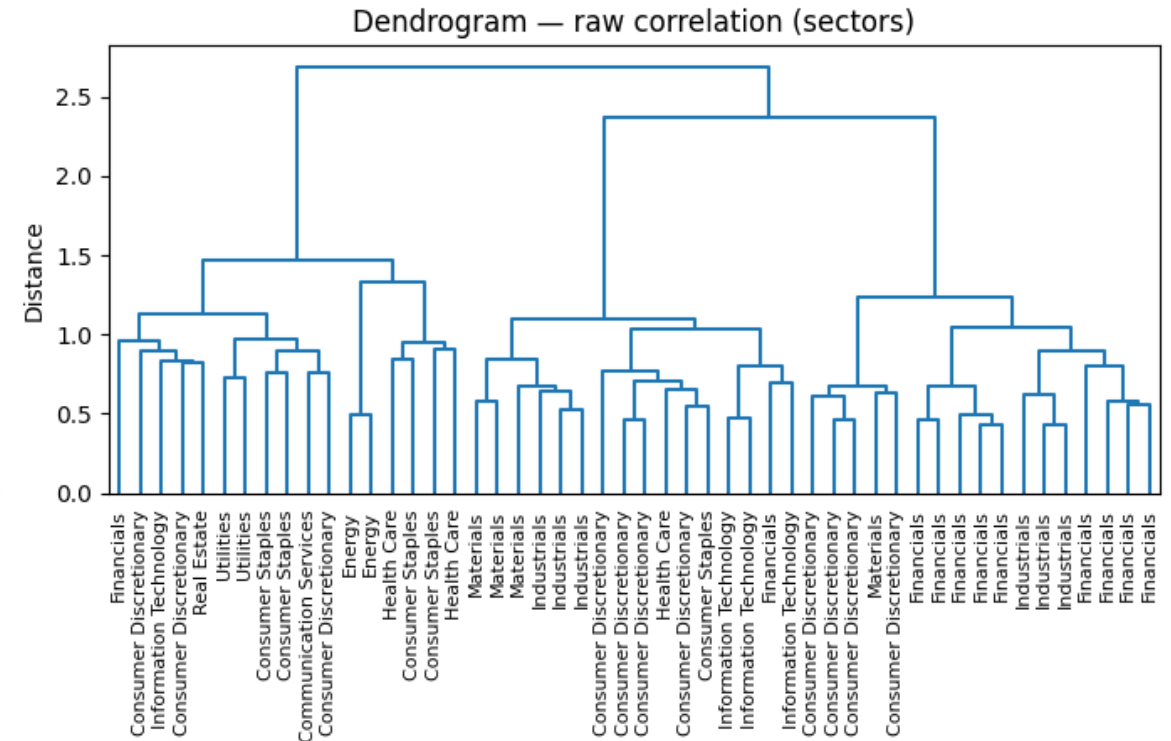
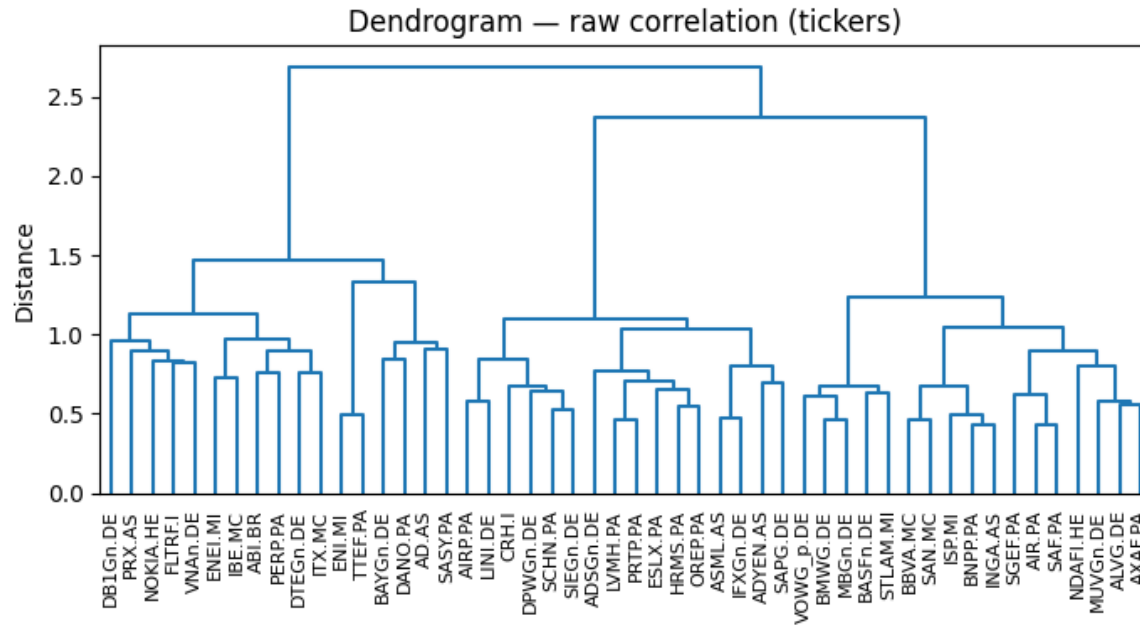
Dendrogram & Cluster Cut

Agglomerative clustering produces a dendrogram whose branch heights encode merge distances. Cutting the tree at any height yields a flat partition of assets into correlation-driven groups — these clusters are the building blocks for HRP.

Key insight: the choice of distance and linkage shapes the clusters — and ultimately the portfolio.

Dendrogram — SX5E, 2023-02-17

Dendrogram — raw correlation | linkage: ward, distance: euclidean | 2023-02-17



Correlation Detoning

Removing the market component before clustering

What Detoning Does

$\tilde{C} = C - u_1 \lambda_1 u_1^T$ (rescale to unit diagonal)

The first eigenvector of an empirical correlation matrix typically captures the *market component* — a common factor that loads roughly equally on every asset.

Detoning subtracts this rank-1 piece (and possibly further systemic eigenvectors), then rescales the diagonal back to one.

Why It Helps Clustering

Like silencing the loudest voice in the room

A pervasive market factor pulls every pair of assets towards positive correlation, washing out the genuine sectoral and idiosyncratic structure clustering tries to recover.

Detoning is the PCA analogue of computing market-adjusted (β -adjusted) returns: it isolates non-market relationships.

Practical Notes

Tone removal: The detoned matrix is singular by construction, so it cannot be plugged into mean–variance optimization — but for clustering, where invertibility is not required, it is exactly the right input.

Key insight: silencing the market reveals the cluster structure that really drives diversification.

05

Hierarchical Risk Parity

López de Prado's three-step recipe for noise-robust allocation

HRP exploits the dendrogram to allocate capital top-down, never inverting Σ . It side-steps Markowitz's Curse — the instability that plagues quadratic optimizers when assets are highly correlated.

The HRP Algorithm

Three stages: cluster, reorder, allocate

1 Tree Clustering

Build the dendrogram from Σ

Convert $\Sigma \rightarrow \rho \rightarrow \text{distance } d_{ij} = \sqrt{0.5(1-\rho_{ij})}$. Apply agglomerative clustering with a chosen linkage to obtain the dendrogram of asset relationships.

2 Quasi-Diagonalization

Reorder Σ by dendrogram leaves

Permute rows and columns of Σ following the dendrogram leaf order. Highly correlated assets cluster around the diagonal — the noisy off-diagonal blocks become visually separated.

3 Recursive Bisection

Top-down split factor $\alpha_k = 1 - \sigma^2_L / (\sigma^2_L + \sigma^2_R)$

Walk the tree from the top, splitting each node into Left and Right sub-clusters. Allocate more capital to the lower-variance side; weights inside each sub-cluster are scaled accordingly.

4 No Matrix Inversion

Cluster variance via inverse-variance parity

Sub-cluster variance is computed using IVP on the cluster's diagonal — no full inverse of Σ . HRP works even when $N > T$, where the sample covariance is singular.

Variants, Benefits & Sensitivities

HRP is a recipe, not a single algorithm

Algorithmic Choices

Many knobs along the pipeline

- Distance metric (correlation, angular, mutual information...)
- Linkage method (single, complete, average, Ward...)
- Cluster variance estimator (IVP, full sub- Σ ...)
- Allocation scheme: top-down recursive bisection vs. full dendrogram iteration

Why Practitioners Like HRP

Stability without giving up structure

- Stability: hierarchical filter dampens covariance noise
- Intuition: capital flows from asset classes $\rightarrow$ sectors $\rightarrow$ names
- No inversion: works when $N > T$ (singular Σ regime)
- Empirically improves out-of-sample performance vs. MVO

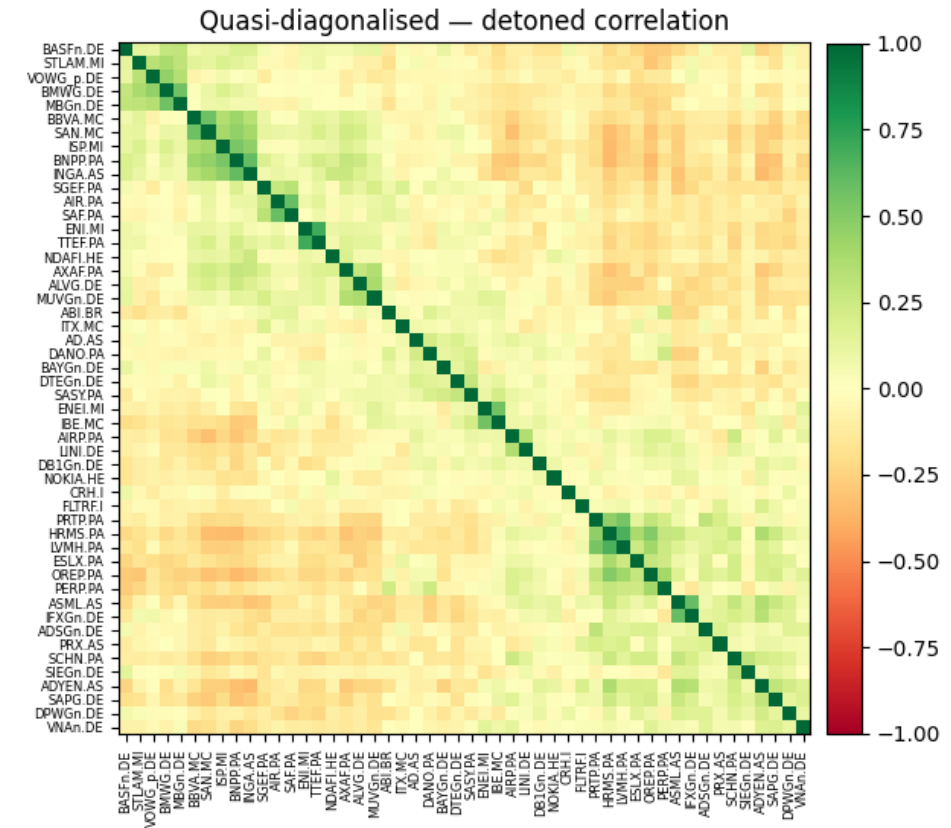
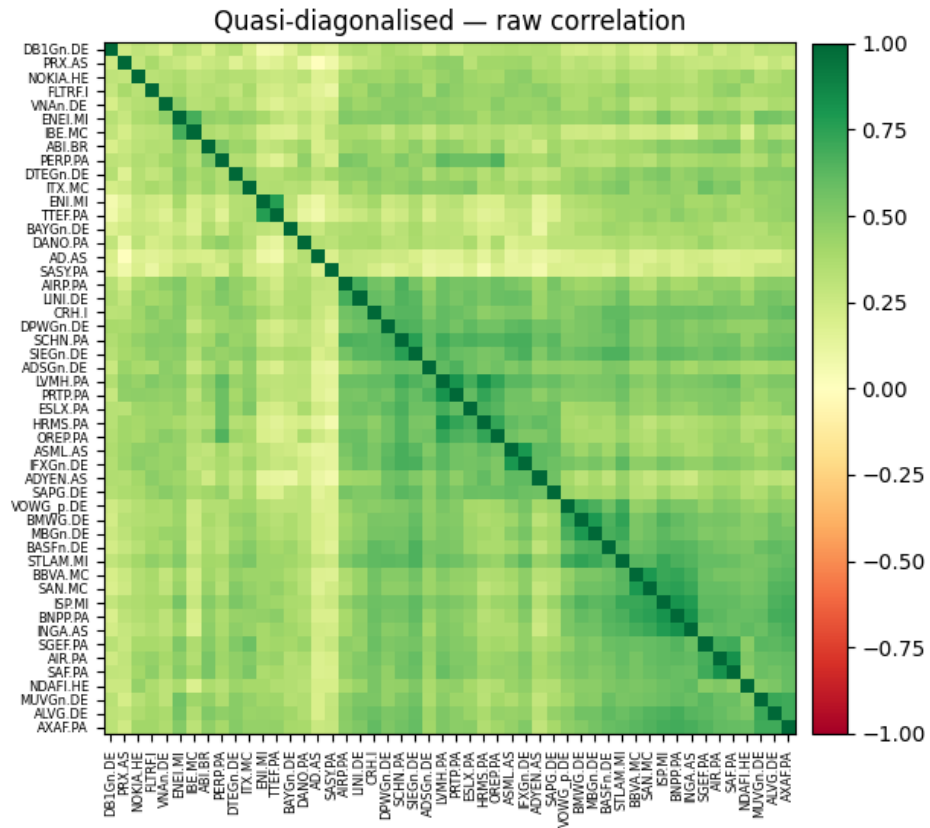
HRP & Markowitz's Curse

Quadratic optimizers treat Σ as a complete graph: every asset talks to every other one. The more correlated the universe, the more unstable the inverse — that's Markowitz's Curse. HRP replaces the complete graph with a hierarchical tree, drastically reducing the effective number of parameters. The trade-off: HRP is still exposed to estimation error in Σ via clustering.

Key insight: structure beats inversion — fewer effective parameters, better out-of-sample stability.

Quasi-Diag Corr Matrix— SX5E, 2023-02-17

Quasi-diagonalised — raw vs detoned | linkage: ward, distance: euclidean | 2023-02-17



References

- [1] Dalio, R., Prince, B., & Jensen, G. (2015). Our Thoughts about Risk Parity and All Weather. Bridgewater Daily Observations.
- [2] Lopez de Prado, M. (2016). Building Diversified Portfolios that Outperform Out of Sample. *Journal of Portfolio Management*, 42(4), 59–69.
- [3] Lopez de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- [4] Lopez de Prado, M. (2020). *Machine Learning for Asset Managers*. Cambridge University Press.
- [5] Maillard, S., Roncalli, T., & Teiletche, J. (2010). On the properties of equally-weighted risk contributions portfolios. *Journal of Portfolio Management*, 36(4), 60–70.

06

Case Study

Polimi Pension Fund — risk-based overlay on the Euro Stoxx 50

A junior analyst studies risk-based allocations on the SX5E constituents. Monthly rebalancing, two-year rolling estimation window, comparing ERC vs HRP — and stress-testing the choices that matter.

What You Will Build

Three parts: ERC, HRP, and the comparison

1 Part I — ERC

Equal Risk Contribution baseline

- Implement an ERC portfolio on the SX5E universe
- Monthly rebalancing
- 2-year rolling estimation window for Σ
- Verify the equal-risk-contribution condition numerically
- Comparison with the inverse-volatility portfolio

2 Part II — HRP

Generic implementation

- Detoning: remove the first PC of ρ before clustering
- Configurable distance metrics & linkage methods
- Two allocation schemes: recursive bisection & dendrogram iteration
- Test cluster stability (Rand index) and shrinkage effects
- HRP weights sensitivity

3 Part III — Comparison

Connecting the dots

- Performance, realized vol, drawdowns, turnover
- ERC vs HRP head-to-head
- Discuss why shrinkage matters less here than in Lab 3

Deliverables & Practical Notes

1

Deliver by Friday 8 May, 18:00 to financial.engineering.polimi@gmail.com — subject "BS: Assignment 4b, Group #".

2

Submit a short PDF with results and discussion.

3

Submit the Python code in your preferred IDE — comment in detail and use explicit variable names.

4

Use the market data in `sx5e_underlyings.csv`.